ICLS – Literature review: viscous inclusions, interface stresses and reactivation of inherited weak zones

Structured review for the ICLS viscous-inclusion work package · 30 October 2026

1 Purpose and protocol

Purpose: place the ICLS results (analytical amplification factors, numerical validation, yield-onset criterion) in the literature and check which parts are known and which may be new.

Review questions

- **RQ1** Analytical solutions for viscous elliptical (2-D) and ellipsoidal (3-D) inclusions; strain-rate and stress partitioning, amplification, closed forms.
- **RQ2** Stress concentration at the inclusion–matrix interface; onset of yielding, failure or shear-zone nucleation at or from inclusions and weak flaws.
- **RQ3** Inherited weak zones in lithosphere-scale models: representation (viscosity, friction/cohesion, plastic strain, grain size, anisotropy), orientation dependence, reactivation criteria.
- **RQ4** Numerical accuracy for inclusion problems: viscosity averaging, convergence, benchmarks.
- **RQ5** 3-D: ellipsoids, finite strike length, comparison of 2-D and 3-D partitioning.

Sources and criteria

About 60 web searches (standard and extended), each review question with several search strings; title checks and citation chasing from key papers (Schmid & Podladchikov 2003; Gilormini & Montheillet 1986; Moulas et al. 2014; Heron et al. 2016); bibliographic data checked on publisher, repository or index pages. Included: peer-reviewed works (and major theses) with analytical or semi-analytical results for linear or power-law viscous inclusions, quantified onset of yielding or localization at weak inhomogeneities, lithosphere models of inherited weak zones with explicit dependence on strength, orientation or geometry, and accuracy studies for inclusion problems. Excluded: purely elastic micromechanics without transfer to viscous flow, field-only studies.

Limitations. No access to Web of Science, Scopus or full citation indices; several publisher pages (Elsevier, Wiley) could not be opened, so some papers were assessed from abstracts and repository records only. The review is systematic in method but not exhaustive; papers marked (abstract only) should be read in full before strong claims of novelty.

2 Findings

RQ1 – Analytical inclusion solutions

- Homogeneous interior fields for ellipsoidal inhomogeneities: Eshelby (1957); transfer to slow viscous flow by Bilby et al. (1975, ellipsoids) and Bilby & Kolbuszewski (1977, elliptical cylinders); early geological theory of strain in inhomogeneous viscous fluids by Gay (1968).
- Gilormini & Montheillet (1986): exact linear-viscous solution for an elliptical (and spheroidal) inclusion, inclusion strain-rate ratios and stresses in the matrix, for plane-strain or axisymmetric tension/compression along the inclusion axes (no oblique orientation); viscoplastic inclusion, power-law matrix by a variational method (full text read).
- Schmid & Podladchikov (2003): complete 2-D solution in general shear (basis of ICLS); Fletcher (2009): anisotropic matrix; Moulas et al. (2014): pressure in and around elliptical inclusions; Halter et al. (2022): finite-difference teaching code for inclusion–matrix systems.
- Power-law and anisotropic generalizations via Eshelby's formalism: Jiang (2013, full text read) gives the complete formalism for viscous ellipsoids (Eshelby tensors from elliptic integrals, a fourth-order strain-rate partitioning tensor relating the ellipsoid's strain rate to the far field,

power-law materials by linearization, numerical implementation), applied to the motion and fabric of deformable ellipsoids; no yield or reactivation analysis. Jiang & Bhandari (2018): pressure deviations (≈ deviatoric stress, consistent with Moulas et al. 2014).

- **Relation to ICLS:** the inclusion values and their dependence on contrast, shape and orientation are classical. The compact formula for $A_{\dot{\varepsilon}}$, the split into contrast and shape parts, the band relation and the symmetry $S(1/\Delta\eta, \beta) = S(\Delta\eta, \beta + 45°)$ were not found in this form; they are reformulations of known results rather than new physics.

RQ2 – Interface stresses and onset of yielding at inclusions

- Gilormini & Montheillet (1986, section 4; full text read) give the interface stress components for loading along the inclusion axes: normal stress σ_{rr} (continuous, uniform in the inclusion), shear traction $\sigma_{r\theta}$ with maximum equal to the inclusion's deviatoric stress (their eq. 37), and the matrix hoop stress $\sigma_{\theta\theta}$ (eqs. 38–41), used for decohesion and matrix cracking (tensile criteria). They do not combine the components into the matrix stress invariant, do not treat oblique orientations, and do not use them for yielding of the matrix. The ICLS statement $\max(A_{\dot{\varepsilon}}, A_{\tau})\cdot\tau_{bg}$ for any orientation is therefore an extension of their result. Their Appendix B notes that the uniform inclusion field requires a linear matrix: after yielding the fields are no longer uniform, which supports restricting the ICLS criterion to the onset.
- Mandal et al. (2001): failure mode of brittle (cohesive) inclusions in a ductile matrix, using Jeffery's viscous solution for the inclusion stress and a Griffith criterion; aspect ratio and orientation control failure.
- Mandal et al. (2004, full text read): PMMA experiments and plane elasticity for a circular, incompressible, purely elastic flaw under a uniaxial far-field stress; the matrix yields (plane-stress von Mises) where the perturbed stress exceeds its strength, first at the flaw tips. The critical bulk stress decreases with the shear-modulus ratio and tends to 0.5 of the matrix yield strength for a hole, the same value as the ICLS onset for a very weak circle ($1/A_{\dot{\varepsilon}} = 1/2$; their loading and yield criterion differ slightly, so the agreement is qualitative). The flaw itself never yields (no strength ratio), only circles are treated, and orientation and shape are not considered. Closest predecessor of the ICLS 'edges yield first' regime; the ICLS criterion extends it to ellipses of any orientation, to yielding of the zone itself and to the regime boundary $\Lambda = \Delta\eta$.
- Experiments and models of strain localization at weak inclusions (Solnhofen limestone in Carrara marble, torsion): strain and stress concentrate in the matrix at the inclusion tips (Nardini et al., 2018, 2020; Döhmann et al., 2019), i.e. the 'edges yield first' regime.
- Weak seeds as nucleation sites of shear bands in visco-plastic models: Kaus (2010). Overpressure in weak shear zones: Schmalholz & Podladchikov (2013).
- **Relation to ICLS:** the ingredients (inclusion stress, interface concentration, yield below matrix strength) exist. Not found: the closed-form onset stress for a viscous elliptical zone with a Drucker–Prager yield stress, the regime boundary $\Lambda = \Delta\eta$ (weak zones) and $\Lambda = 1$ (stiff zones) independent of shape and orientation, and its use for inherited weak zones.

RQ3 – Inherited weak zones in lithosphere models

- Representations: reduced friction angle in a Drucker–Prager material (Heron et al., 2016; LaMEM multi-IWZ setup of this project); plastic strain softening of cohesion and friction (Xue et al., 2024); damage/strain-dependent weakening memory (Fuchs & Becker, 2019, 2022); compositional and rheological components of former orogens (Petersen & Schiffer, 2016; Salazar-Mora et al., 2018); offset weak zones (Chenin & Beaumont, 2013); viscous anisotropy (Tommasi et al., 2009; analytical strain-rate enhancement of an anisotropic layer: Liu et al., 2024); heterogeneities in transpression (Jourdon et al., 2026).
- Orientation: weaknesses perpendicular to extension are most readily reactivated in analogue models (Zwaan et al., 2021); weak-zone strength and obliquity control rift segmentation in 3-D models (Xue et al., 2024); frictional reactivation of planes follows Sibson (1985), as summarized for rift basins by Samsu et al. (2023).
- **Relation to ICLS:** orientation and strength effects are studied numerically or experimentally case by case; no analytical criterion that combines the viscous contrast with the plastic

strength contrast was found. The ICLS point that an isotropic weak volume has no frictional optimum orientation (orientation enters only through the viscous amplification) was not found stated explicitly.

RQ4 – Numerical accuracy

- First-order convergence with viscosity jumps inside cells (Duretz et al., 2011); averaging and Eulerian/Lagrangian comparisons (Deubelbeiss & Kaus, 2008); harmonic averaging best for velocity and pressure errors of a strong circular inclusion (Thielmann et al., 2014).
- **Relation to ICLS:** consistent. Not found: that the inclusion means behave differently from the bulk errors (harmonic averaging fails for the means of stiff inclusions although the bulk errors are smallest), and the sensitivity of component-based stresses to vertex viscosities.

RQ5 – 3-D

- Eshelby (1957), Bilby et al. (1975) and the generalized Eshelby approach of Jiang (2013) give the 3-D ellipsoid solutions needed for the planned 3-D counterpart; Jiang's strain-rate partitioning tensor is the direct 3-D analogue of $A_{\dot{\varepsilon}}$ and can be evaluated with his published procedure; no study was found that maps the 2-D to 3-D reduction of amplification for finite strike length in the context of inherited weak zones.

3 Novelty assessment

ICLS result	Closest prior work	Status
Uniform inclusion fields, inclusion values vs. Δη, a/b, θ	Eshelby 1957; Bilby et al. 1975, 1977; Gilormini & Montheillet 1986; Schmid & Podladchikov 2003	known
Compact $A_{\dot{\varepsilon}}$, contrast × shape split, band relation, S(1/Δη, β) = S(Δη, β+45°)	implicit in the above	new formulation of known results
Matrix maximum on the interface = $\max(A_{\dot{\varepsilon}}, A_{\tau})\cdot\tau_{bg}$	Gilormini & Montheillet 1986: interface components for loading along the axes (max shear traction = inclusion stress)	components known; invariant for any orientation and use for matrix yielding not found
Yield onset $\tau_{bg}^{*}/\tau_{y,m}$, regime boundary $\Lambda = \Delta\eta / \Lambda = 1$	Mandal et al. 2004 (matrix yielding at elastic circular flaws; hole limit 0.5 as in ICLS); Mandal et al. 2001	edges regime for circles known; zone yielding, regime boundary, shape and orientation not found
No frictional optimum for an isotropic weak zone; β = 45° optimal through $A_{\dot{\varepsilon}}$	Sibson 1985 (planes); Zwaan et al. 2021; Xue et al. 2024	not found stated explicitly
Inclusion means vs. bulk errors for averaging schemes	Deubelbeiss & Kaus 2008; Duretz et al. 2011; Thielmann et al. 2014	minor addition
Application to the LaMEM IWZ parameters (Λ from c, φ; β = dip)	Heron et al. 2016 (same representation)	new application

Read in full: Gilormini & Montheillet (1986), Mandal et al. (2004), Jiang (2013). Still open: the citing literature of Schmid & Podladchikov (2003) and Mandal et al. (2004).

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